Namrota, 5th November 2026

BEHIND THE SCENES OF INSTAGRAM

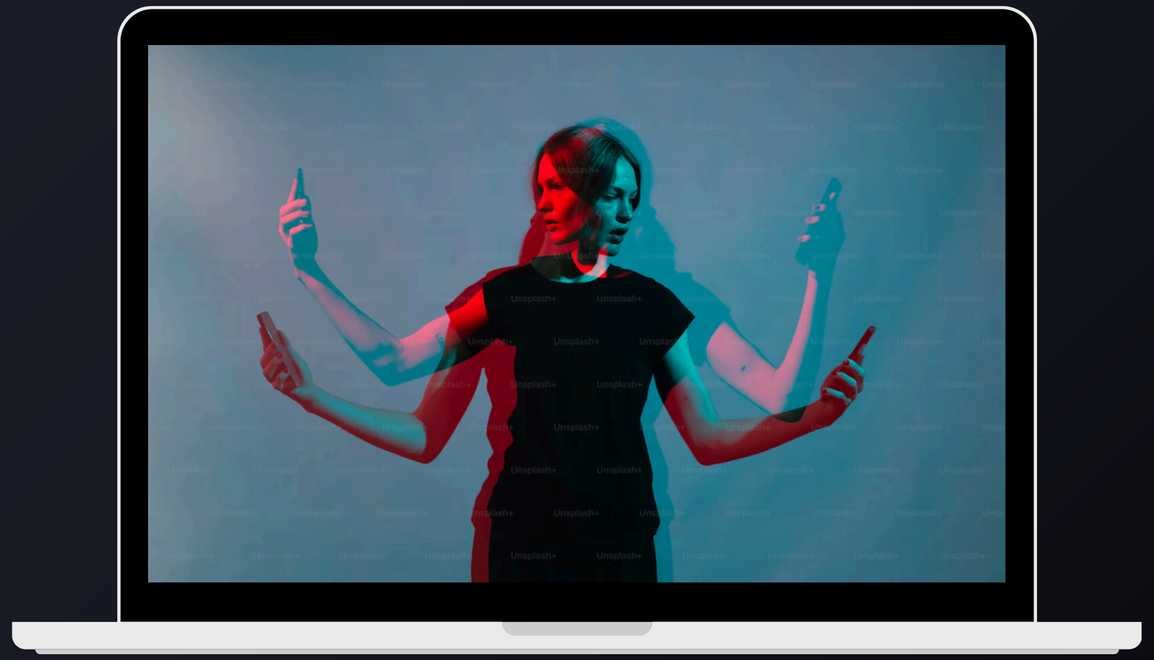




How data is stored, distributed and processed

Namrota

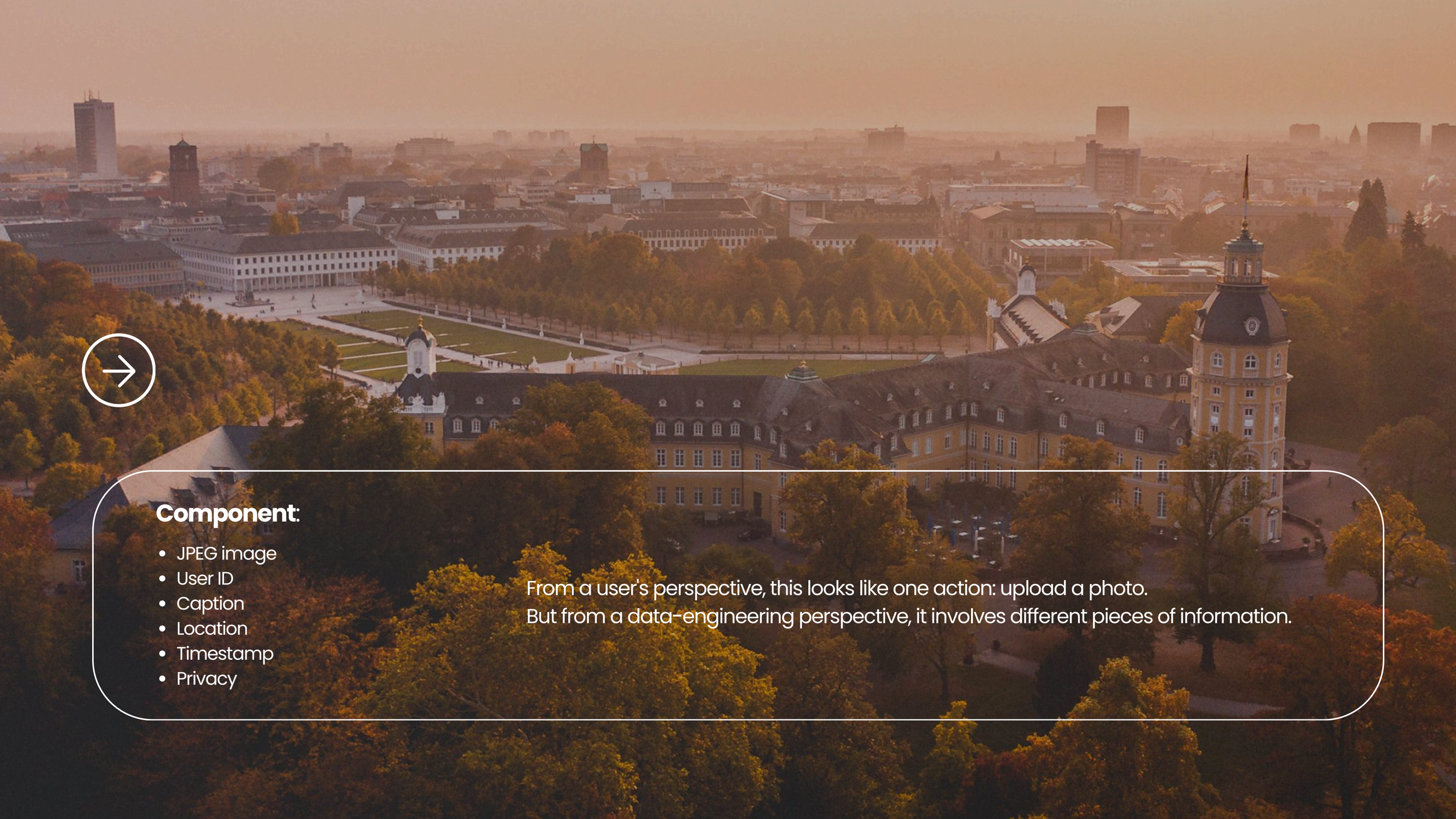
WHAT HAPPENS WHEN SOMEONE UPLOADS A POST?



Social media platforms manage several types of data, and each type has different access patterns.

At its core Instagram is a social graph- users, posts, relationships, interactions, and activities.

- user profiles
- photos and videos
- comments and likes
- direct messages
- stories and reels
- follows and relationships
- activity feed and notifications



Component:

- JPEG image
- User ID
- Caption
- Location
- Timestamp
- Privacy

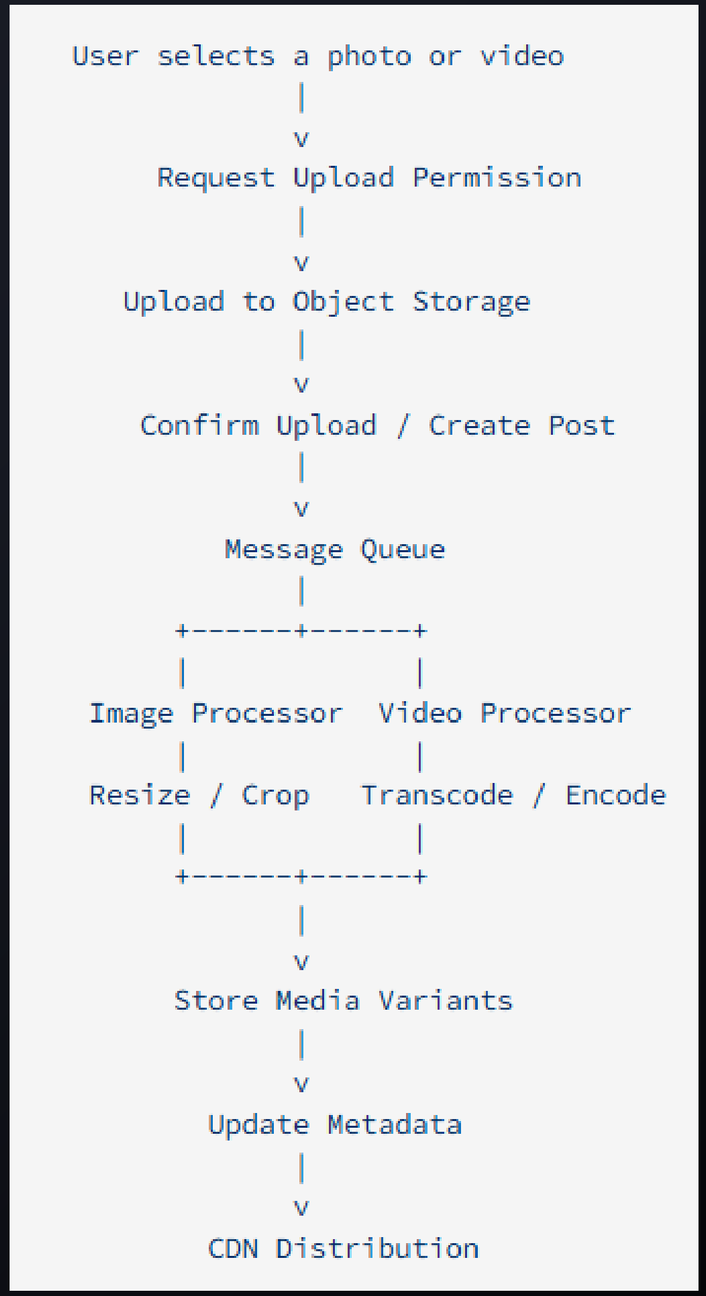
From a user's perspective, this looks like one action: upload a photo.
But from a data-engineering perspective, it involves different pieces of information.

Instagram client prepares upload

A. Media file
B. Post Information

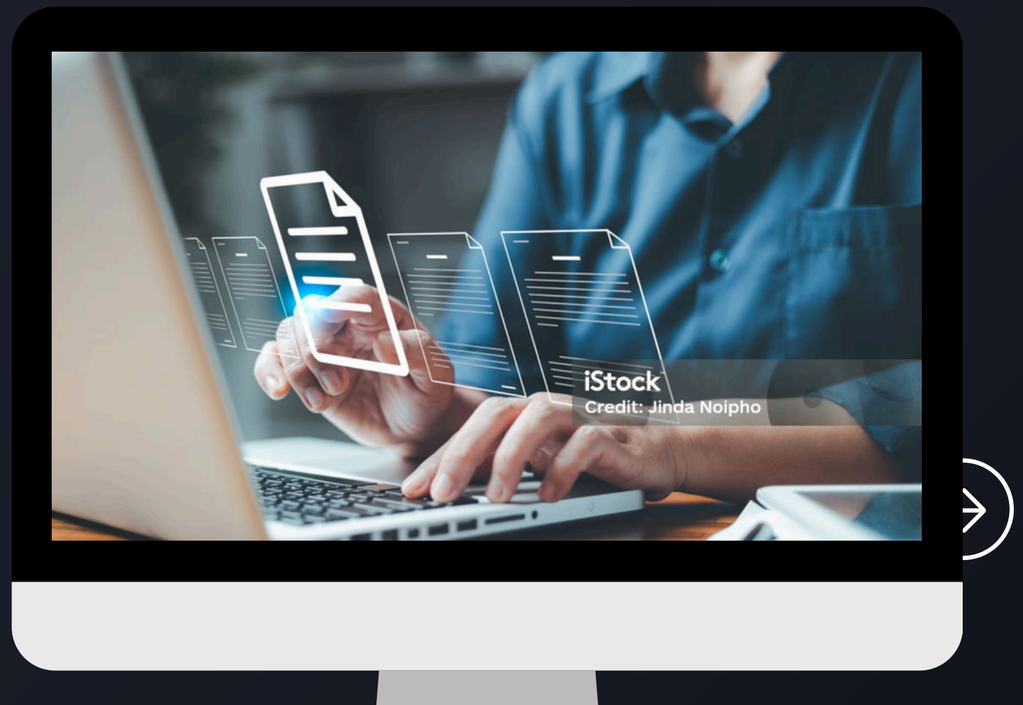
An illustrative example regarding post information:

```
{
 "user_id": 101,
 "caption": "Beautiful sunset",
 "location": "Karlsruhe",
 "visibility": "followers_only",
 "media_type": "image/jpeg"
}
```



MAKING THE POST PERSISTENT

Once the media upload has been accepted, the platform also needs a persistent post record




```
CREATE TABLE posts (
    post_id BIGINT PRIMARY KEY,
    user_id BIGINT NOT NULL,
    caption TEXT,
    visibility VARCHAR(30),
    status VARCHAR(20),
    created_at TIMESTAMP
);

CREATE TABLE media (
    media_id BIGINT PRIMARY KEY,
    post_id BIGINT NOT NULL,
    object_key VARCHAR(500) NOT NULL,
    media_type VARCHAR(50),
    FOREIGN KEY (post_id)
        REFERENCES posts(post_id)
);
```

THANK YOU

